

Equal-Determinant Coherence: Sharp Fiber Phase Envelopes, a Conditional Positive-Coherence Gate, and a Census Schema

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Abstract

We isolate the first genuinely arithmetic obstruction behind determinant-energy survival in the fixed-shift matched cutoff shell. The literal coefficient is written as a sum of native triplet weights u_t over normalized-determinant fibers F_n . For prescribed fiber moduli r_i , we prove that the complete interval of attainable resultant magnitudes in the unrestricted free-phase comparison class is

$$[(2r_{\max} - \sum_i r_i)_+, \sum_i r_i].$$

This gives the sharp phase-blind lower envelope

$$\sum_n (2r_{\max, n} - R_n)_+$$

for the post-bin ℓ^1 mass. It can vanish even when the raw absolute mass is natural, so a raw lower bound and a fiber-survival theorem are logically distinct. The extremizing free phases need not be realizable by the coupled phases of the literal physical packet.

We then prove the exact identity

$$D = E_{\text{trip}} + C_{\text{eq}}$$

between determinant energy, the native triplet diagonal, and the ordered off-diagonal equal-determinant correlation. The literal TPC-32 packet bounds, imported on the high- β physical domain $C = \lfloor J \rfloor \ll Q$, imply

$$E_{\text{trip}} \ll_{\epsilon} X^{\epsilon} J Q^2 \asymp X^{-1+\epsilon} Q^3 J^2.$$

Hence any energy lower bound with fixed loss smaller than 1 requires a positive off-diagonal term of asymptotically the same size. This is a conditional positive-coherence gate: no such lower bound is proved here.

Finally, we give an exponent ledger and an audit-ready proposed native-key census schema. A raw-mass loss λ_R followed by a fiber-survival loss λ_{fib} certifies energy loss $2(\lambda_R + \lambda_{\text{fib}})$ under the verified $O(Q)$ support bound. Candidate duplicate, symmetry, and shared-divisor pairings are classified without assigning them an unproved main term. The proposed implementation aggregates exact determinant keys by streaming or external sort and obtains $C_{\text{eq}} = D - E_{\text{trip}}$ without enumerating all pairs. No census is executed and no archive or data are supplied. The finite identities close an algebraic L0 gate; the TPC-32 diagonal comparison, the derived conditional coherence gate, and the proposed schema are L1 interfaces, not the missing growing-scale L2 arithmetic lower bound.

Keywords: fixed shift; determinant fiber; phase envelope; coherent energy; prime-Möbius packet; arithmetic census; endpoint audit.

Convention. Every arithmetic statement concerns the literal matched coefficient at one fixed nonzero shift h_0 , with complete native keys, actual coefficients and active support, and the original global normalization. A lower bound for a raw absolute majorant is never reported as a lower bound after determinant aggregation. Every TPC-32 estimate imported below is restricted to its high- β theorem domain $C = \lfloor J \rfloor \ll Q$.

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1 Introduction

The matched-shell determinant coefficient of TPC-32 is a signed sum. TPC-83 shows that its natural determinant energy is equivalent, at subpower resolution, to survival of the *post-bin* ℓ^1 mass [4, 6]. That equivalence is exact and useful, but it moves rather than removes the next obstruction: why should literal contributions landing in the same determinant bin fail to cancel?

This paper resolves the finite-dimensional part of that question and then states the remaining arithmetic task in an audit-ready form. There are two separate losses:

1. the literal native packet may have too little raw absolute mass;
2. even if that mass is natural, equal-determinant aggregation may cancel it.

Neither loss is controlled by the upper-bound axioms presently available. In particular, bounded row multipliers are allowed to vanish, and all native coefficients can be rescaled down while preserving those axioms.

1.1 The three conclusions

Our first conclusion is a sharp phase-envelope theorem. In one fiber, knowledge of the moduli alone forces a nonzero resultant exactly when one term dominates the sum of all the others. Summed over fibers this produces a computable, optimal lower certificate for the post-bin mass. If this certificate vanishes on most of the raw mass, the failure is conclusive for every phase-blind argument, but says nothing against a signed arithmetic mechanism.

Our second conclusion is structural. The determinant energy is not primarily a diagonal quantity at its natural scale. If

$$N_0 = JQ^2, \quad QJ \asymp X,$$

then the natural energy is

$$\frac{N_0^2}{Q} = Q^3 J^2,$$

whereas the literal triplet diagonal is at most

$$X^{o(1)} JQ^2 \asymp X^{-1+o(1)} Q^3 J^2.$$

Thus a successful energy theorem with any fixed loss $\lambda_D < 1$ would have to prove *positive aggregate coherence* among distinct native terms with equal determinant. The present paper proves this conditional L1 necessity statement, not its energy-lower-bound hypothesis.

Our third conclusion is methodological. We specify an audit-ready schema for a future native archive and census. The proposed classification distinguishes literal diagonal terms, exact duplicate atoms, determinant-preserving symmetries, shared-target/shared-divisor pairings, and an unpaired remainder. No census is executed here, and no asymptotic size is assigned to the last four classes without exact key data and a proof.

1.2 What is new and what is not

The phase envelope, the energy decomposition, and the conditional mass-to-energy transfer are finite L0 results. The diagonal upper bound and support statement are L1 physical imports from TPC-32, used only in its high- β domain $C = \lfloor J \rfloor \ll Q$. Together they convert a vague “cancellation problem” into a conditional signed lower-correlation gate. The proposed archive and census schema are an L1 interface, not a completed computation.

The missing L2 theorem is still substantial: one must prove a growing-scale lower bound for the actual raw mass, the actual post-bin mass, or a positive family of off-diagonal equal-determinant pairings. General results on multiplicative functions, including short-interval and logarithmically averaged theorems [2, 3], do not automatically apply to the restricted high-row geometry and its literal weights.

1.3 Fixed-shift and endpoint discipline

All statements are made at one fixed $h_0 \neq 0$. No averaging over shifts, determinant frequencies, row slopes, or masks may replace the physical coefficient. The full route also has a strict endpoint budget: every accumulated fixed-power loss must be

$$\boxed{\text{strictly smaller than } 1/400.}$$

Equality is not reserve and is a stopping point. This paper neither identifies h_0 with 2 nor asserts any prime-pair consequence. The determinant-energy interface and the physical endpoint loss are kept as independent ledgers; an energy exponent is not automatically added to the latter without a proved crosswalk.

1.4 Organization

[Section 2](#) fixes the literal objects. [Section 3](#) proves the sharp fiber theorem. [Section 4](#) transfers raw and fiber losses to energy. [Section 5](#) isolates the conditional positive-coherence gate. [Section 6](#) opens the Möbius variables and marks the Chowla boundary. [Sections 7](#) and [8](#) classify and specify the proposed census schema. [Sections 9](#) and [10](#) give success, kill, and claim rules.

2 Literal native packet and determinant fibers

Fix $h_0 \in \mathbb{Z} \setminus \{0\}$. We use the one-packet notation of TPC-32 and suppress X from symbols when no ambiguity is possible. The row and orbit scales obey

$$QJ \asymp_{\mathscr{D}} X, \quad N_0 := JQ^2, \quad (1)$$

and the content cutoff is $C = \lfloor J \rfloor$. We work only in the inherited high- β regime

$$J \ll_{\mathscr{D}} Q, \quad C = \lfloor J \rfloor \ll_{\mathscr{D}} Q, \quad (2)$$

which is the theorem domain required by the TPC-32 estimates imported below. No proposition in this paper imports those estimates outside (2). The symbol $\mathscr{D}$ collects fixed supports, residue conditions, smooth weights, and positive margins.

The symbol R below denotes raw absolute mass. When the inherited divisor coefficient b_R is used in [section 6](#), its subscript denotes the typed TPC-32 sieve cutoff R_{cut} , not the raw mass. We retain the historical notation b_R only to make comparison with TPC-32 immediate.

2.1 Native triplets

A native triplet is the physical key

$$t = (\alpha, \gamma, j)$$

after the three shell channels have been recombined pointwise. Its archive record also stores content, orientation, and coefficient provenance, but these metadata do not create additional copies of t . Its row targets are

$$N_{\alpha}(j) = m_{\alpha}j + h_0, \quad N_{\gamma}(j) = m_{\gamma}j + h_0.$$

The normalized determinant label is

$$\nu(t) = \Delta_{\alpha, \gamma}^{\#}(j) := \frac{m_{\alpha} - m_{\gamma}}{(N_{\alpha}(j), N_{\gamma}(j))} \in \mathbb{Z} \setminus \{0\}. \quad (3)$$

All channel terms are first recombined into the exact matched shell. At that level a convenient triplet weight is

$$u_t = \gamma_{\alpha}^{(1)} \gamma_{\gamma}^{(2)} \mathfrak{A}_{\alpha, \gamma}(j) \mathcal{K}_{\alpha, \gamma}^{\text{sh}}(j) \mathbf{1}_{c_t \leq C}, \quad c_t = (N_{\alpha}(j), N_{\gamma}(j)). \quad (4)$$

If an implementation opens channel, cutoff, divisor, or provenance fields further, every expansion record retains its parent-triplet identifier and literal aggregation multiplier. Expansion records are not additional native triplets. This prevents a formal opening from silently changing E_{trip} , multiplicity, or normalization.

Definition 2.1 (Fibers and masses). Let $\mathcal{T}$ be the finite multiset of native keys, retaining distinct keys even if their displayed numerical fields happen to coincide. For $n \in \mathbb{Z}$, put

$$F_n = \{t \in \mathcal{T} : \nu(t) = n\}, \quad A(n) = \sum_{t \in F_n} u_t. \quad (5)$$

Define

$$R = \sum_{t \in \mathcal{T}} |u_t|, \quad M = \sum_n |A(n)|, \quad (6)$$

$$D = \sum_n |A(n)|^2, \quad E_{\text{trip}} = \sum_{t \in \mathcal{T}} |u_t|^2. \quad (7)$$

The exact active support is

$$S = \#\{n : A(n) \neq 0\}.$$

The distinction between R and M is fundamental. The triangle inequality gives

$$0 \leq M \leq R, \quad (8)$$

but gives no reverse inequality. The global normalizer in (1) is not divided by the number of fibers, channels, or native records.

2.2 Inherited upper bounds

Proposition 2.2 (L1 imported one-sided information). *For the literal TPC-32 packet in (2) at $C = \lfloor J \rfloor$,*

$$R \ll_{\varepsilon, h_0, \mathcal{D}} X^\varepsilon N_0, \quad (9)$$

$$M \leq R, \quad (10)$$

$$S \ll_{\mathcal{D}} Q, \quad (11)$$

$$D \ll_{\varepsilon, h_0, \mathcal{D}} X^\varepsilon Q^3 J^2. \quad (12)$$

None of these statements is a lower bound for R or M .

Proof. The TPC-32 physical absolute-mass interface bounds the sum of the absolute values of complete matched summands by $X^\varepsilon J Q^2$. This gives (9); then (10) is (8). The determinant labels lie in an interval of length $O_{\mathcal{D}}(Q)$, giving (11). Finally, the inherited singleton bound and raw absolute mass yield (12); equivalently one may use the post-bin ℓ^∞ -by- ℓ^1 estimate from TPC-32 [4, 6]. $\square$

Remark 2.3 (Why the lower bound is absent). The permitted joint multiplier $\mathfrak{A}_{\alpha, \gamma}(j)$ is only bounded above and may vanish. Multiplying every row or mask coefficient by X^{-B} preserves all stated upper axioms while making R arbitrarily small on the natural scale. Therefore no uniform lower bound for R , M , or D follows from those axioms alone.

3 Sharp phase envelopes inside determinant fibers

The first question is purely geometric. How much of a fiber's absolute mass survives if only its moduli are known?

Theorem 3.1 (Complete resultant interval). *Let $m \geq 1$, let $r_1, \dots, r_m \geq 0$, and set*

$$R_f = \sum_{i=1}^m r_i, \quad r_{\max} = \max_{1 \leq i \leq m} r_i,$$

and allow arbitrary phases $\theta_i \in \mathbb{R}$. Then

$$\left\{ \left| \sum_{i=1}^m r_i e^{i\theta_i} \right| : \theta_1, \dots, \theta_m \in \mathbb{R} \right\} = [(2r_{\max} - R_f)_+, R_f]. \quad (13)$$

Both endpoints, and every magnitude between them, are attained.

Proof. We prove by induction that the attainable magnitudes form the stated closed interval. The assertion is immediate for $m = 1$.

Suppose the first $m - 1$ lengths have attainable interval $[L', R']$, where

$$R' = \sum_{i < m} r_i, \quad L' = (2r'_{\max} - R')_+.$$

Write $r = r_m$. If a partial resultant has magnitude s , adding the last vector produces every magnitude in $[|s - r|, s + r]$, by varying the angle between the two vectors. As s runs continuously through $[L', R']$, these intervals overlap and their union is an interval. Its upper endpoint is $R' + r$, and its lower endpoint is the distance from r to $[L', R']$:

$$\min_{s \in [L', R']} |s - r| = \max\{L' - r, r - R', 0\}. \quad (14)$$

If $r \geq r'_{\max}$, the right side of (14) equals $(2r - (R' + r))_+$. If $r < r'_{\max}$, then $L' - r = (2r'_{\max} - (R' + r))_+$ whenever it is positive, while $r - R' \leq 0$ because $R' \geq r'_{\max} > r$. Thus in both cases the new lower endpoint is

$$(2 \max\{r'_{\max}, r\} - (R' + r))_+.$$

This is exactly the claimed formula for m lengths. The induction also proves that every intermediate magnitude is attained. $\square$

Remark 3.2 (Polygon interpretation). The lower endpoint is zero precisely when the longest segment is no longer than the sum of the remaining segments. This is the polygon closure condition. When one segment dominates, placing all other segments directly opposite it attains the positive defect $2r_{\max} - R_f$. The theorem is the sharp form of the polygon inequality [1].

For the actual fiber F_n , define

$$R_n = \sum_{t \in F_n} |u_t|, \quad r_{\max, n} = \max_{t \in F_n} |u_t|, \quad d_n = (2r_{\max, n} - R_n)_+, \quad (15)$$

with all three quantities zero for an empty fiber.

Corollary 3.3 (Sharp global magnitude certificate). *For every choice of phases consistent with the fiber moduli,*

$$M = \sum_n |A(n)| \geq M_{\min}^{\text{mag}} := \sum_n d_n. \quad (16)$$

Moreover,

$$M_{\min}^{\text{mag}} \leq M \leq R = \sum_n R_n, \quad (17)$$

and both global endpoints are attainable by independent choices of phases within the unrestricted free-phase comparison class. Hence no lower bound depending only on the multiset of fiber moduli can improve (16).

Proof. Apply theorem 3.1 separately to every nonempty fiber. The determinant fibers are disjoint, so their phases may be chosen independently in the free-phase comparison class. Choosing a minimizing configuration in every fiber attains the sum of lower endpoints; aligning every fiber attains $\sum_n R_n = R$. $\square$

Remark 3.4 (Free-phase sharpness versus physical realizability). The lower bound (16) applies to the literal coefficient because it uses only its actual moduli. The endpoint attainability statement is sharper in a different sense: it is optimal over all complex packets having the same fiber partition and moduli while allowing the phases to vary independently. Literal phases may be coupled through shared rows, masks, smooth factors, or orientations, so an extremizing free-phase assignment need not be realizable by the physical packet. The corollary therefore proves optimality for modulus-only arguments, not the existence of a physical cancellation or coherence configuration.

Definition 3.5 (Raw and fiber survival). When $R > 0$, define the actual fiber-survival ratio and its phase-blind certificate by

$$\vartheta_{\text{fib}} = \frac{M}{R}, \quad \vartheta_{\text{mag}} = \frac{M_{\min}^{\text{mag}}}{R}. \quad (18)$$

Then $0 \leq \vartheta_{\text{mag}} \leq \vartheta_{\text{fib}} \leq 1$.

Remark 3.6 (What a zero envelope means). It is possible that $d_n = 0$ in every fiber while R is arbitrarily large. Thus a zero phase-blind certificate kills the *magnitude-only route*, not the actual signed coefficient. A positive arithmetic pairing can still force a large resultant. The proposed census schema in [section 8](#) is designed to distinguish these two outcomes in a future executed audit.

Example 3.7 (Sharpness at both extremes). *If every fiber is a singleton, then $d_n = R_n$, so $M = R$ independently of phase. If one fiber contains two equal moduli, then its lower envelope is zero and opposite phases attain complete cancellation. No interpolation inequality can distinguish those cases without the actual fiber multiplicities and moduli.*

4 The raw–fiber–energy loss ledger

We now translate the two survival stages into the determinant-energy exponent needed by the fixed-shift route. The notation $X^{-\lambda+o(1)}$ records a fixed loss exponent λ ; all soft factors are taken only after the fixed margins have been chosen.

Definition 4.1 (Two losses). A *raw nondegeneracy certificate* of loss $\lambda_R \geq 0$ is

$$R \geq X^{-\lambda_R+o(1)} N_0. \quad (19)$$

A *fiber-survival certificate* of loss $\lambda_{\text{fib}} \geq 0$ is

$$M \geq X^{-\lambda_{\text{fib}}+o(1)} R. \quad (20)$$

The second certificate may come either from the sharp magnitude envelope

$$M_{\min}^{\text{mag}} \geq X^{-\lambda_{\text{fib}}+o(1)} R$$

or from a stronger theorem using the actual signed arithmetic.

Theorem 4.2 (Certified energy loss). *Assume (19) and (20). If the actual postaggregation support satisfies*

$$S \leq X^{\lambda_S+o(1)} Q, \quad (21)$$

then

$$D \geq X^{-\lambda_D^{\text{cert}}+o(1)} Q^3 J^2, \quad \lambda_D^{\text{cert}} = 2\lambda_R + 2\lambda_{\text{fib}} + \lambda_S. \quad (22)$$

For the verified TPC-32 support bound $S \ll Q$, this specializes to

$$\boxed{\lambda_D^{\text{cert}} = 2(\lambda_R + \lambda_{\text{fib}})}. \quad (23)$$

Proof. Cauchy–Schwarz on the actual nonzero determinant bins gives

$$M^2 = \left(\sum_{A(n) \neq 0} |A(n)| \right)^2 \leq S \sum_n |A(n)|^2 = SD.$$

The two survival hypotheses imply

$$M \geq X^{-(\lambda_R + \lambda_{\text{fib}})+o(1)} N_0.$$

Together with (21),

$$D \geq \frac{M^2}{S} \geq X^{-2\lambda_R - 2\lambda_{\text{fib}} - \lambda_S + o(1)} \frac{N_0^2}{Q}.$$

Since $N_0 = JQ^2$,

$$\frac{N_0^2}{Q} = Q^3 J^2,$$

which proves the theorem. $\square$

Remark 4.3 (Why the fiber loss is doubled). The exponent $2\lambda_{\text{fib}}$ is not an artifact of an estimate: energy is quadratic in the surviving post-bin mass. The equal-amplitude construction in TPC-83 makes the M^2/S inequality sharp, so no universal improvement is possible without using more than mass and support [6].

4.1 Compatibility with a zero-mode gain

Suppose a later transfer theorem supplies a fixed zero-mode gain η_Z and requires

$$\lambda_D \leq 2\eta_Z. \quad (24)$$

Then the present route is sufficient provided

$$\lambda_R + \lambda_{\text{fib}} + \frac{\lambda_S}{2} \leq \eta_Z. \quad (25)$$

With the literal $S \ll Q$ support bound, this becomes

$$\lambda_R + \lambda_{\text{fib}} \leq \eta_Z.$$

Equality in (25) leaves no reserve for any other fixed-power loss. It is therefore an interface equality, not a successful end-to-end budget.

4.2 The inherited reverse direction

The literal post-bin singleton bound at $C = \lfloor J \rfloor$ gives

$$L := \max_n |A(n)| \ll X^{o(1)} \frac{N_0}{Q}. \quad (26)$$

Because $D \leq LM$, any certificate

$$D \geq X^{-\lambda_D + o(1)} \frac{N_0^2}{Q} \quad (27)$$

forces

$$M \geq X^{-\lambda_D + o(1)} N_0. \quad (28)$$

Together with [theorem 4.2](#), this recovers the sharp TPC-83 interval: a post-bin mass loss λ_M certifies energy loss $2\lambda_M$, whereas energy loss λ_D forces post-bin mass loss no worse than λ_D . Neither direction provides the missing lower bound by itself.

4.3 Independent energy and physical endpoint ledgers

The determinant-energy certificate and the end-to-end physical loss are separate accounting objects. The first must meet the transfer interface

$$\boxed{\lambda_D^{\text{cert}} \leq 2\eta_Z.} \quad (29)$$

Let Λ_{phys} denote the sum of fixed-power degradations that a complete physical transfer theorem proves must be paid, with no duplicated slack and no absorption into $o(1)$. Independently, the endpoint requires

$$\boxed{\Lambda_{\text{phys}} < \frac{1}{400}.} \quad (30)$$

We do not assert any formula that automatically adds λ_D^{cert} to previously incurred physical losses. Such a formula would require a term-by-term crosswalk proving that a determinant-energy loss survives as a new physical degradation rather than already being encoded in the zero-mode interface or another transfer factor. Without that crosswalk, adding the two ledgers may double count.

Equality in (30) is a stop. Equality in (29) meets that interface but leaves no energy reserve and is not by itself end-to-end success. Averaging over h , frequencies, or alternative packets does not repair a failure of either ledger for the literal fixed- h_0 coefficient.

5 The conditional positive-coherence gate

The quadratic energy can be opened before any estimate. Define the ordered off-diagonal equal-determinant correlation

$$C_{\text{eq}} := \sum_n \sum_{\substack{t, t' \in F_n \\ t \neq t'}} u_t \overline{u_{t'}}. \quad (31)$$

The word “ordered” means that both (t, t') and (t', t) occur.

Proposition 5.1 (Exact diagonal-coherence identity). *For every finite native packet,*

$$\boxed{D = E_{\text{trip}} + C_{\text{eq}}.} \quad (32)$$

Moreover $C_{\text{eq}} \in \mathbb{R}$, although it need not be nonnegative.

Proof. Opening one fiber gives

$$|A(n)|^2 = \sum_{t, t' \in F_n} u_t \overline{u_{t'}} = \sum_{t \in F_n} |u_t|^2 + \sum_{\substack{t, t' \in F_n \\ t \neq t'}} u_t \overline{u_{t'}}.$$

Sum over n . The off-diagonal ordered terms occur in conjugate pairs, so their total is real. $\square$

5.1 The imported diagonal upper bound loses at least one power

Proposition 5.2 (L1 imported triplet-diagonal upper bound). *For the literal TPC-32 matched packet in the high- β domain (2),*

$$E_{\text{trip}} \ll_{\varepsilon, h_0, \mathcal{D}} X^\varepsilon J Q^2 = X^\varepsilon N_0. \quad (33)$$

Consequently,

$$\boxed{E_{\text{trip}} \ll_{\varepsilon, h_0, \mathcal{D}} X^{-1+\varepsilon} Q^3 J^2.} \quad (34)$$

Proof. From (4), the joint multiplier is uniformly bounded, and the exact prefix envelope gives

$$|\mathcal{K}_{\alpha, \gamma}^{\text{sh}}(j)| \ll_{\varepsilon, \mathcal{D}} X^\varepsilon$$

on the physical support. That support contains $O_{\mathcal{D}}(J)$ orbit integers. Hence, after starting with a smaller soft exponent,

$$\begin{aligned} E_{\text{trip}} &\ll X^\varepsilon \sum_j \sum_{\alpha, \gamma} |\gamma_\alpha^{(1)}|^2 |\gamma_\gamma^{(2)}|^2 \\ &\ll X^\varepsilon J \left(\sum_\alpha |\gamma_\alpha^{(1)}|^2 \right) \left(\sum_\gamma |\gamma_\gamma^{(2)}|^2 \right). \end{aligned}$$

The row energy bounds

$$\sum_\alpha |\gamma_\alpha^{(i)}|^2 \ll_{\mathcal{D}} Q \log(2L), \quad i = 1, 2,$$

give (33) after absorbing logarithms into X^ε . Finally, $QJ \asymp X$ implies

$$Q^3 J^2 = (QJ)(JQ^2) \asymp X N_0,$$

which yields (34). $\square$

Theorem 5.3 (L1 conditional positive coherence below loss one). *Fix $0 \leq \lambda_D < 1$. If along a growing sequence*

$$D \geq X^{-\lambda_D + o(1)} Q^3 J^2, \quad (35)$$

then

$$C_{\text{eq}} = D - E_{\text{trip}} = (1 - o(1))D. \quad (36)$$

In particular $C_{\text{eq}} > 0$ eventually and has the same fixed-power scale as D .

Proof. Choose $\varepsilon > 0$ strictly smaller than $1 - \lambda_D$. By [proposition 5.2](#),

$$\frac{E_{\text{trip}}}{D} \ll X^{-1+\varepsilon+\lambda_D+o(1)} = o(1).$$

Now use [\(32\)](#). □

Corollary 5.4 (L1 conditional off-diagonal target). *At subpower loss, $D \geq X^{-o(1)}Q^3J^2$ can hold only if*

$$C_{\text{eq}} \geq X^{-o(1)}Q^3J^2.$$

Thus an argument proving cancellation of the entire off-diagonal sum down to $o(Q^3J^2)$ cannot prove natural determinant energy.

Remark 5.5 (The decorrelation trap). The precise hypothesis that makes the diagonal asymptotic is

$$C_{\text{eq}} = o(E_{\text{trip}}).$$

Under this hypothesis, $D = (1 + o(1))E_{\text{trip}} \ll X^{o(1)}N_0$. Since $Q^3J^2 \asymp XN_0$, this lies at most on the loss-one scale and therefore loses at least one full power, up to soft factors. The conditional route to a lower bound below loss one is not “more cancellation everywhere”; it must prove that a specific structured part of [\(31\)](#) is positive and macroscopic.

Remark 5.6 (A distinct energy from TPC-33). The column upper-energy $E_L + E_R$ used in the zero-frequency analysis of TPC-33 is not the determinant energy D [\[5\]](#). An upper bound for that column energy cannot be substituted for the lower bound in [\(35\)](#).

6 Opening the arithmetic and locating the correlation boundary

The exact matched shell is a difference of two quadratic prefixes. For $Y \in \{U_0, T\}$, write

$$\epsilon_{U_0} = 1, \quad \epsilon_T = -1,$$

and recall

$$b_R(w) = -\mu(w)w_R(w)$$

on its squarefree support, with $w_R(w) > 0$. The native weight may be opened as

$$\begin{aligned} u_t = & \sum_{Y \in \{U_0, T\}} \epsilon_Y \gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \mathfrak{A}_{\alpha, \gamma}(j) \\ & \times \sum_{\substack{u, v \leq Y \\ u|N_\alpha(j), v|N_\gamma(j)}} b_R(u)b_R(v) \mathbf{1}_{c_t \leq C}. \end{aligned} \tag{37}$$

Every cutoff sign, row orientation, content restriction, and aggregation multiplier is part of the native reconstruction record, but its expansion children remain attached to the same parent triplet.

6.1 Four Möbius factors after crossing

The row provenance contains $\mu(d_\alpha)$ and $\mu(d_\gamma)$. Target primitivity gives $(d_\alpha, u) = (d_\gamma, v) = 1$ on contributing terms, so on the two legs

$$\mu(d_\alpha)b_R(u) = -\mu(d_\alpha u)w_R(u), \quad \mu(d_\gamma)b_R(v) = -\mu(d_\gamma v)w_R(v). \tag{38}$$

Squaring [\(37\)](#), or crossing two distinct native atoms in one determinant fiber, therefore produces structured correlations containing four Möbius factors, together with positive divisor weights and literal smooth, residue, cutoff, and mask coefficients. Schematically, one encounters

$$\mu(d_\alpha u)\mu(d_\gamma v)\mu(d_{\alpha'} u')\mu(d_{\gamma'} v')\mathbf{1}_{\nu(t)=\nu(t')}, \tag{39}$$

not an unweighted generic four-point correlation.

Remark 6.1 (Do not over-fuse). Any Möbius factor introduced by exact-content inversion may share a prime factor with u or v . It cannot be fused into [\(38\)](#) unless the requisite coprimality has been proved for the literal key. A future census must store such factors separately.

6.2 The Chowla boundary

A theorem uniform over a broad affine family must be tested against its smallest specializations. For example, if its admissible parameters include $a = s = h = 1$, so that consecutive unit-scale targets are allowed, then a specialization can contain

$$\sum_{t \leq x} \mu(t) \mu(t+1), \quad (40)$$

the classical unweighted two-point Chowla correlation. An unconditional theorem uniform enough to imply cancellation in (40) would include a currently unproved special case. Logarithmically averaged variants are known [3], but they are not literal substitutes for the unweighted high-row packet.

This observation is a stress test, not an equivalence theorem. The actual family here has large structured rows, fixed support restrictions, determinant equality, divisor weights, and coupled targets. We do *not* prove that its correlation problem is equivalent to classical Chowla. A localization, completion, or pairing mechanism may exploit precisely those restrictions.

6.3 Why a positive term is plausible but unproved

Equation (39) shows two competing features:

1. generic off-diagonal terms have genuinely oscillatory Möbius signs;
2. exact repeated targets, divisors, or native atoms can pair those signs and leave positive weights.

Theorem 5.3 says conditionally that the second feature would have to dominate at the desired scale. It does not say which pairings occur or how many survive. The proposed schema organizes the arithmetic census gate still missing.

Proposition 6.2 (No lower bound from the opened formula alone). *The representation (37), together with the current boundedness and support axioms, does not imply a positive fixed-power lower bound for R , M , or C_{eq} .*

Proof. As in remark 2.3, the allowed literal multipliers can vanish or be uniformly rescaled. Moreover, (37) is a signed difference between two cutoff levels. An identity and upper envelopes for its summands cannot exclude cancellation within a native atom, within a determinant fiber, or across the ordered correlation sum. $\square$

7 A literal classification of candidate positive pairings

The identity $D = E_{\text{trip}} + C_{\text{eq}}$ makes candidate pairings worth examining only after determinant equality and literal coefficients have been checked. The following hierarchy prevents the same pair from being counted twice. It is a classification protocol, not an asymptotic theorem.

The proposed precedence is explicit: after removing the diagonal, Class 1 (proved exact duplicate) takes priority over Class 2 (a registered determinant-preserving symmetry), which takes priority over Class 3 (a registered shared-target or shared-divisor relation); every remaining ordered pair is Class 4. A future executable census must freeze this precedence and the finite lists of allowed symmetry and shared-relation tests in a versioned registry. No such machine-readable registry is supplied here, so the class totals below are proposed L1 outputs, not computed results.

7.1 Class 0: the literal diagonal

Pairs with $t = t'$ contribute exactly

$$E_{\text{trip}} = \sum_t |u_t|^2.$$

They are nonnegative, and the imported upper bound places their total at least one full power below natural determinant energy, up to soft factors, by proposition 5.2. Treating an expanded channel or divisor record as a new triplet would artificially inflate this class; identity is determined by the parent triplet (α, γ, j) , not by a row in a flattened expansion table.

7.2 Class 1: exact duplicate physical atoms

Distinct native keys $t \neq t'$ are *exact duplicates* only if the archive proves equality of the complete physical atom data:

$$\nu(t) = \nu(t'), \quad u_t = u_{t'}, \quad (41)$$

including cutoff sign, row coefficients, mask, orientation, content, divisor provenance, and aggregation multiplier. Such an ordered pair contributes $|u_t|^2 \geq 0$, and the contribution is strictly positive only when $u_t \neq 0$. A numerical equality after floating-point rounding is not a proof of (41).

If a physical atom occurs with literal multiplicity m , its within-copy energy is $m^2|u|^2$: $m|u|^2$ is diagonal and $m(m-1)|u|^2$ is nonnegative off-diagonal coherence. Conversely, records created solely by a computational expansion do not acquire physical multiplicity.

7.3 Class 2: determinant-preserving symmetries

A map $\sigma : t \mapsto \sigma t$ is a candidate symmetry pairing only after proving

$$\nu(\sigma t) = \nu(t) \quad (42)$$

and tracking the exact coefficient ratio

$$u_{\sigma t}/u_t.$$

Positivity requires the real part of $u_t \overline{u_{\sigma t}}$ to have a controlled positive sign after summation.

The most tempting permutation fails the first test. Exchanging the two rows gives

$$\Delta_{\gamma, \alpha}^{\#}(j) = -\Delta_{\alpha, \gamma}^{\#}(j). \quad (43)$$

Since the off-diagonal mask removes determinant 0, plain row exchange maps F_n to F_{-n} , not F_n , and contributes nothing to C_{eq} as defined. It becomes relevant only if an additional archived transformation restores the determinant sign and preserves all other physical orientations.

7.4 Class 3: shared-target or shared-divisor pairings

Two distinct keys may satisfy exact relations such as

$$N_{\alpha}(j) = N_{\alpha'}(j'), \quad N_{\gamma}(j) = N_{\gamma'}(j'), \quad (44)$$

or may share selected divisor variables after (37) is expanded. Such equalities can pair Möbius factors and leave positive weights. They are candidates only if all of the following are verified:

1. the exact target or divisor equations;
2. the determinant identity $\nu(t) = \nu(t')$;
3. the literal cutoff signs and orientations;
4. the true multiplicity after inverse aggregation;
5. the sign of the complete coefficient product.

Sharing a target residue class, a content value, or a divisor bound is not the same as sharing the literal target or divisor.

7.5 Class 4: genuinely unpaired remainder

After Classes 1–3 are removed in the declared order above, all remaining ordered pairs form the unpaired remainder. Its total may be positive or negative. Bounding each term absolutely is generally too expensive; proving uniform cancellation over an enlarged affine family may cross the Chowla boundary described in section 6. A viable theorem may instead localize the remainder by content, determinant multiplicity, or a geometry-specific completion.

Proposition 7.1 (What the classification proves). *For any conjugation-closed priority partition of the ordered equal-determinant pairs,*

$$C_{\text{eq}} = C_{\text{dup}} + C_{\text{sym}} + C_{\text{share}} + C_{\text{rem}}. \quad (45)$$

The duplicate term is nonnegative when (41) is proved. No sign or asymptotic size for the other three terms follows from the classification alone.

Proof. The equality is a disjoint partition of the finite ordered sum (31). Conjugation closure makes every class sum real. Exact duplicates have $u_t \overline{u_{t'}} = |u_t|^2$. The remaining assertions require additional arithmetic information not contained in their labels. $\square$

Remark 7.2 (A positive main-term route). A successful alternative to the mass ledger would identify a conjugation-closed family $\mathcal{P}$ with

$$\sum_{(t,t') \in \mathcal{P}} u_t \overline{u_{t'}} \geq X^{-\lambda_P + o(1)} Q^3 J^2$$

and prove that the complement is smaller. This is a positive-pairing theorem, not an assertion that every off-diagonal pair is positive.

8 An audit-ready proposed native-key census schema

The proposed computation is not a search over rounded determinant bins. This section specifies an audit-ready schema and a scalable execution design for a future census of the literal coefficient. No census is executed in this paper, and no native archive, parameter manifest, machine-readable schema, code, or census output is supplied. Thus the word “proposed” is essential: the requirements below are not themselves a reproducibility claim. Every listed field is needed either to reconstruct u_t , to prove determinant equality, or to prevent a change of normalization.

8.1 Archive schema

For each fixed X , fixed h_0 , and fixed packet, a future implementation should archive:

1. **Experiment identity:** code revision, parameter file, fixed h_0 , smooth-support identifiers, residue data, and random-seed field if any. The typed packet-scale tuple is
$$(L, D_{\text{row}}, Q, J, R_{\text{cut}}, T, U_0, C).$$
2. **Native row keys:** complete α, γ , row slopes m_α, m_γ , row provenance, literal complex coefficients $\gamma_\alpha^{(1)}, \gamma_\gamma^{(2)}$, and row orientation.
3. **Orbit and targets:** j , exact integers $N_\alpha(j), N_\gamma(j)$, their gcd c_t , and every active support flag.
4. **Determinant data:** numerator $m_\alpha - m_\gamma$, exact signed quotient $\nu_t = (m_\alpha - m_\gamma)/c_t$, and a proof flag for exact divisibility.
5. **Expansion-tree data:** parent native-triplet key, matched-shell channel, $Y \in \{U_0, T\}$, cutoff sign ϵ_Y , orientation label, and inverse aggregation map. These child records do not enlarge the native triplet multiset.
6. **Divisor expansion data:** exact u, v , divisor witnesses, $b_R(u), b_R(v)$, squarefree and coprimality flags, and any exact-content inversion variable kept as a separate field.
7. **Joint multiplier:** literal $\mathfrak{A}_{\alpha, \gamma}(j)$, its mask, periodic and smooth factors, and a provenance hash.
8. **Aggregation data:** physical multiplicity, computational expansion multiplicity, aggregation multiplier, and the final exact or certified-interval value of u_t .

The complete native key is a tuple of these identity fields. A future implementation must fix a canonical serialization, hash algorithm, schema version, and precision policy. A hash then supplies a reproducibility check, not a mathematical proof of equality. None of those machine-readable choices is frozen in this paper.

8.2 Fiber statistics

After exact aggregation, the proposed output records for every determinant n :

$$\begin{aligned} m_n &= \#F_n, & R_n &= \sum_{t \in F_n} |u_t|, \\ r_{\max, n} &= \max_{t \in F_n} |u_t|, & d_n &= (2r_{\max, n} - R_n)_+, \\ A(n) &= \sum_{t \in F_n} u_t, & D_n &= |A(n)|^2. \end{aligned} \tag{46}$$

Global checks are

$$\begin{aligned} R &= \sum_n R_n, & M_{\min}^{\text{mag}} &= \sum_n d_n, & M &= \sum_n |A(n)|, \\ D &= \sum_n D_n, & E_{\text{trip}} &= \sum_t |u_t|^2, & C_{\text{eq}} &= D - E_{\text{trip}}. \end{aligned} \tag{47}$$

8.3 Scalable determinant aggregation

Let $N_{\text{rec}} = \#\mathcal{T}$ be the number of native records after inverse aggregation to parent triplets, and let B be an available-memory record budget. The proposed exact-key pipeline is:

1. stream the expansion records, verify inverse multiplicities, aggregate them to native parent keys, emit $(\nu(t), t, u_t)$, and accumulate E_{trip} ;
2. stable-sort the emitted records by the exact integer determinant key $\nu(t)$, using bounded in-memory runs and an external merge when $N_{\text{rec}} > B$;
3. scan each determinant fiber once to compute $m_n, R_n, r_{\max, n}, A(n), D_n$, and then sum R, M, D ;
4. compute the total off-diagonal correlation by the exact identity

$$C_{\text{eq}} = D - E_{\text{trip}},$$

rather than by forming all ordered pairs.

With comparison sorting this requires $O(N_{\text{rec}} \log N_{\text{rec}})$ comparisons, $O(N_{\text{rec}})$ disk storage, and $O(B)$ working memory, apart from the standard external-merge pass count. When the exact keys fit in memory, deterministic sorting has the same comparison bound and hash aggregation has expected $O(N_{\text{rec}})$ work. Exact or directed-interval accumulators must be used throughout; the final subtraction returns an exact value or a certified interval.

8.4 Compressed class joins

Class totals should also avoid a global pair table. Exact duplicates are obtained by sorting or hashing the complete physical-atom key and using group sums and multiplicities. For a fixed registry of K determinant-preserving maps, symmetry candidates are obtained by K indexed key lookups. Shared-target and shared-divisor candidates are produced by exact sort-merge joins on their declared integer relation keys. Their nominal costs are, respectively,

$$O(N_{\text{rec}} \log N_{\text{rec}}), \quad O(K N_{\text{rec}} \log N_{\text{rec}}), \quad O(N_{\text{rec}} \log N_{\text{rec}} + Z),$$

where Z is the join-output size; hash indexes may replace the logarithmic lookup factors in expected time.

The quantity Z can be quadratic. Whenever a class contribution factorizes over a relation-key group, the implementation must store compressed group sums, squared norms, counts, and formally verified inverse multiplicities instead of materializing its pairs. The unpaired remainder is then computed by subtracting the disjoint certified class totals from C_{eq} . A full ordered-pair table is permitted only for a declared bounded exploratory run. A class whose relation does not admit certified compression must report its output-sensitive cost and is not called scalable merely because the global determinant aggregation is scalable.

8.5 Classification fields

Every pair materialized in a bounded run, and every compressed class join representing such pairs, carries:

1. an exact equal-determinant witness;
2. a class label from [section 7](#) and the declared priority-rule and registry versions;
3. duplicate, symmetry, shared-target, and shared-divisor flags, each accompanied by the exact equations used;
4. coefficient product $u_t \overline{u_{t'}}$, or a rigorous complex interval enclosing it;
5. physical multiplicity and expansion multiplicity separately;
6. a conjugate-partner identifier, ensuring each reported class is real;
7. the contribution to the chosen candidate main term and to its complement.

8.6 Exactness and support rules

Integer targets, gcds, divisors, determinants, and Möbius signs are computed exactly. Rational algebraic coefficients are stored exactly when possible; otherwise directed interval arithmetic must certify all signs and inequalities used in a theorem. Floating-point phase proximity is exploratory evidence only. Overlapping numerical intervals do not prove exact coefficient equality: Class 1 requires a symbolic identity, a shared canonical provenance proof, or another exact equality certificate. Otherwise the pair remains outside the duplicate class.

The active support is

$$\{n : A(n) \neq 0\}$$

after complete aggregation. It may not be replaced by the available determinant interval, a thresholded support, or the set of raw fiber labels. If numerical precision cannot decide whether $A(n) = 0$, the bin is marked unresolved and no exact support count is reported.

8.7 Scaling tests

For a growing sequence of X , a future census should report the nonnegative loss diagnostics

$$\lambda_R(X) = \frac{[\log(N_0/R)]_+}{\log X}, \quad (48)$$

$$\lambda_{\text{fib}}(X) = \frac{[\log(R/M)]_+}{\log X}, \quad (49)$$

$$\lambda_D(X) = \frac{[\log(Q^3 J^2/D)]_+}{\log X}, \quad (50)$$

where $[y]_+ = \max\{y, 0\}$, whenever the displayed ratios are defined. The report also includes the untruncated raw ratios, uncertainty intervals, and local slopes. These quantities are diagnostics, not asymptotic exponents until a theorem controls their limits.

The following residual checks detect normalization errors:

$$0 \leq M_{\min}^{\text{mag}} \leq M \leq R, \quad \frac{M^2}{S} \leq D \leq LM, \quad D - E_{\text{trip}} = C_{\text{eq}}. \quad (51)$$

where the middle lower bound is asserted only when $A \neq 0$. Every future run must record the original global $N_0 = JQ^2$; no per-fiber or per-triplet renormalization is permitted.

Remark 8.1 (Sampling is not a completed census). A sampled pair table can estimate which classes deserve attention, but it cannot prove multiplicity, support, or a signed remainder bound. A publishable L1 computation may use the compressed exact joins above, provided the serialization, registry, compression map, inverse multiplicities, manifests, and output checksums are versioned and formally verified. Those implementation artifacts remain future work.

9 Success certificates, partial advances, and kill criteria

This section turns the preceding identities into a branchable research plan. The purpose is to make a failed route informative without converting it into a claim about the whole parity problem.

9.1 Route A: raw mass followed by fiber survival

Route A succeeds at losses $(\lambda_R, \lambda_{\text{fib}}, \lambda_S)$ only if all of the following are proved for the literal fixed- h_0 packet:

$$R \geq X^{-\lambda_R+o(1)} N_0, \quad (52)$$

$$M \geq X^{-\lambda_{\text{fib}}+o(1)} R, \quad (53)$$

$$S \leq X^{\lambda_S+o(1)} Q. \quad (54)$$

Then

$$\lambda_D^{\text{cert}} = 2\lambda_R + 2\lambda_{\text{fib}} + \lambda_S.$$

The verified support theorem supplies $\lambda_S = 0$. If a zero-mode transfer has gain η_Z , require

$$\lambda_D^{\text{cert}} \leq 2\eta_Z,$$

and separately verify the strict complete loss budget (30). The certified energy loss is not automatically added to Λ_{phys} ; such an addition requires the crosswalk described after (30).

The fiber step may use the sharp magnitude certificate:

$$\sum_n (2r_{\text{max},n} - R_n)_+ \geq X^{-\lambda_{\text{fib}}+o(1)} R.$$

If that fails, Route A remains open to a signed arithmetic lower bound for M .

9.2 Route B: a positive pairing main term

Choose a rigorously defined conjugation-closed pair family $\mathcal{P}$ and write

$$C_{\text{eq}} = C_{\mathcal{P}} + C_{\mathcal{R}}.$$

Route B succeeds if

$$C_{\mathcal{P}} \geq X^{-\lambda_P+o(1)} Q^3 J^2, \quad (55)$$

$$|C_{\mathcal{R}}| = o(C_{\mathcal{P}}), \quad (56)$$

or if a one-sided remainder estimate leaves a positive constant fraction of $C_{\mathcal{P}}$. Then

$$D = E_{\text{trip}} + C_{\mathcal{P}} + C_{\mathcal{R}} \geq X^{-\lambda_P+o(1)} Q^3 J^2.$$

The candidate family must preserve the exact determinant and all physical orientations. A pairing main term found only after averaging over h or changing the packet is not a Route B certificate.

9.3 Publishable partial advances

The following remain useful even when neither route reaches the final endpoint:

1. a rigorous positive raw exponent λ_R ;
2. a nontrivial magnitude-survival exponent for a positive share of the raw mass;
3. an exact multiplicity theorem for one pairing class;
4. a positive $C_{\mathcal{P}}$ main term with a remainder too large for the endpoint;

5. a localization theorem showing that a broad Chowla-hard remainder is confined to a small, explicitly weighted class;
6. an exact negative census showing that a named symmetry or duplicate mechanism has insufficient mass.

Each result must retain its L0/L1/L2 label and measured loss. A positive exponent larger than the available budget is progress on the local gate, not completion of the physical route.

9.4 Kill criteria

A branch stops, or is relabeled purely exploratory, if any of the following occurs:

1. it proves only upper bounds for R or M ;
2. the sharp magnitude envelope is zero on all but a negligible share of the raw mass, and the branch has no signed mechanism;
3. a proposed pairing changes determinant sign, target, orientation, cutoff sign, or physical multiplicity;
4. its gain appears only after averaging the fixed shift, determinant frequency, row slopes, masks, or incompatible packets;
5. the required remainder theorem, uniformly stated, contains an unproved Chowla specialization and no geometry-specific localization;
6. the certified energy loss violates $\lambda_D^{\text{cert}} \leq 2\eta_Z$;
7. the complete physical loss is at least $1/400$, including equality.

A killed branch supports only a branch-local negative conclusion: for example, “magnitude dominance does not certify fiber survival in this packet.” It does not prove a universal parity barrier or rule out a different fixed-shift mechanism.

9.5 Next theorem target

The most informative next theorem is one of the following:

- (i) a raw nondegeneracy lower bound for R ;
- (ii) a census theorem showing $\sum_n d_n \geq X^{-\lambda_{\text{fib}} + o(1)} R$;
- (iii) an exact positive main term from duplicate or shared-target/shared-divisor pairings;
- (iv) a localization of the unpaired remainder to a family on which an existing mean-value theorem applies without introducing an unproved special case.

A future executed census should decide which of (ii)–(iv) has enough literal mass before a long analytic argument is attempted. The schema in [section 8](#) does not itself make that decision.

10 Claim boundary and conclusion

10.1 Level ledger

The results of this paper have three deliberately separate levels:

L0: exact algebra and finite geometry.

The complete resultant interval, the sharp global magnitude envelope, the raw–fiber–energy exponent transfer, and $D = E_{\text{trip}} + C_{\text{eq}}$ are rigorous finite-dimensional results.

L1: typed physical interface.

On the high- β domain $C = \lfloor J \rfloor \ll Q$, the imported TPC-32 support, singleton, and triplet-diagonal bounds put E_{trip} at most on the loss-one scale, so it loses at least one power relative to natural energy, up to soft factors; the conditional positive-coherence gate derived from that import is also L1. The native-key schema, scalable aggregation design, pairing classification, invariants, and success/kill criteria specify an audit-ready proposed interface for the actual fixed-shift packet. No census has been executed and no archive or data are supplied.

L2: growing arithmetic theorem.

A uniform lower bound for the literal R , M , or positive C_{eq} at growing X is not proved. This remains the next arithmetic gate.

10.2 What is proved

Magnitude data have exactly one universal way to force fiber survival: a dominant native term. Its sharp contribution is $(2r_{\max,n} - R_n)_+$. Separately, if an energy theorem below loss one is proved, the imported diagonal upper bound forces its main scale to come from positive off-diagonal equal-determinant coherence. These two facts sharply narrow the search: either enough fibers are magnitude-dominant, or a structured signed pairing must be found and its remainder controlled.

10.3 What is not proved

We do not prove:

- a lower bound for the actual raw or post-bin mass;
- a positive asymptotic for any unverified pairing class;
- a fixed-2 estimate or a theorem uniform in a way that can be specialized to it;
- unweighted two-point Chowla;
- a breach of the parity barrier;
- the Hardy–Littlewood prime-pair asymptotic;
- infinitude of twin primes.

No numerical census, regardless of range, upgrades an L1 pattern to an L2 theorem without uniform error control.

10.4 Conclusion

TPC-83 identifies post-bin mass as the energy-survival variable. The present paper opens that variable one level deeper. It proves the optimal phase-blind certificate and isolates a conditional positive-coherence gate: any future determinant-energy lower bound with loss below one would force a same-scale positive off-diagonal correlation. It does not prove that lower bound or exhibit such a correlation. The proposed audit schema concentrates the remaining uncertainty in a finite list of literal pairing mechanisms and a precisely defined unpaired remainder, but those class totals have not been computed.

This is forward progress in the sense appropriate to the fixed-shift program: an algebraic ambiguity has been closed, an arithmetic gate has been isolated, and every route through that gate carries an explicit exponent and endpoint audit. It is not a prime-pair conclusion.

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